



September 2024 Exam

EC400

Dynamic Programming and Differential Equations

Suitable for all candidates

Instructions to candidates

This exam contains ten questions. Answer all questions.

If during this exam you feel you need to make additional assumptions to proceed, then please do so stating your assumptions clearly.

Time Allowed	<i>90 minutes</i>
You are supplied with:	<i>No additional materials</i>
You may also use:	<i>No additional materials</i>
Calculators:	<i>Calculators are not allowed in this exam</i>

EC400: Exam

Dmitry Mukhin

d.mukhin@lse.ac.uk

Consider a problem of an entrepreneur with an initial stock of capital K_0 choosing a path of consumption and capital $\{C_t, K_{t+1}\}_{t=0}^{\infty}$ to maximize utility

$$\sum_{t=0}^{\infty} \beta^t \log C_t$$

subject to a resource constraint

$$K_{t+1} = K_t^\alpha - C_t,$$

where $\beta, \alpha \in (0, 1)$ are given parameters.

1. (10 marks) Write down the Lagrangian and take the first-order conditions.
2. (8 marks) Derive the intertemporal optimality condition (Euler equation). Using analogy with a finite-horizon model, write down the transversality condition.
3. (10 marks) Interpret the intertemporal optimality condition using the perturbation argument, i.e. a small deviation from the optimal path.
4. (8 marks) Solve for the (interior) steady state of the economy.
5. (6 marks) Write down the Bellman equation for the problem.
6. (14 marks) Conjecture that $V(X) = A \log X + B$, where X is the state variable and A, B are unknown constants. Solve the Bellman equation, find the values of A and B and the policy function.
7. (6 marks) Note that the resource constraint can be rewritten for an arbitrary length of period Δ as $K_{t+\Delta} - K_t = (K_t^\alpha - K_t - C_t)\Delta$, where $K_t\Delta$ corresponds to capital depreciation. Derive the continuous time version of the resource constraint.

8. (12 marks) Define the discount factor ρ such that $\beta = e^{-\rho}$. Write down the Hamiltonian and the optimality conditions according to the maximum principle.
9. (12 marks) Derive the continuous-time version of the Euler equation and provide an economic interpretation.
10. (14 marks) Write down the HJB equation and use the first-order and envelope conditions to obtain the Euler equation.